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- Published: date not stated
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](<https://cdn.elie.net/static/files/text-based-captcha-strengths-and-weaknesses/text-based-captcha-strengths-and-weaknesses-slides.pdf>) | | Conference | Computer and Communications

Security (CCS) - 2011 | | | Authors | Elie Bursztein , Matthieu Martin , John C. Mitchell show more | | Citation | Bibtex | |

Warning:the attack techniques and design guidelines described in this paper are obsolete due to the invention of [generalized techniques to break text captchas as described in this paper](#).

We carry out a systematic study of existing visual CAPTCHAs based on distorted characters that are augmented with anti-segmentation techniques. Applying a systematic evaluation methodology to 15 current CAPTCHA schemes from popular web sites, we find that 13 are vulnerable to automated attacks. Based on this evaluation, we identify a series of recommendations for CAPTCHA designers and attackers, and possible future directions for producing more reliable human/computer distinguishers.

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